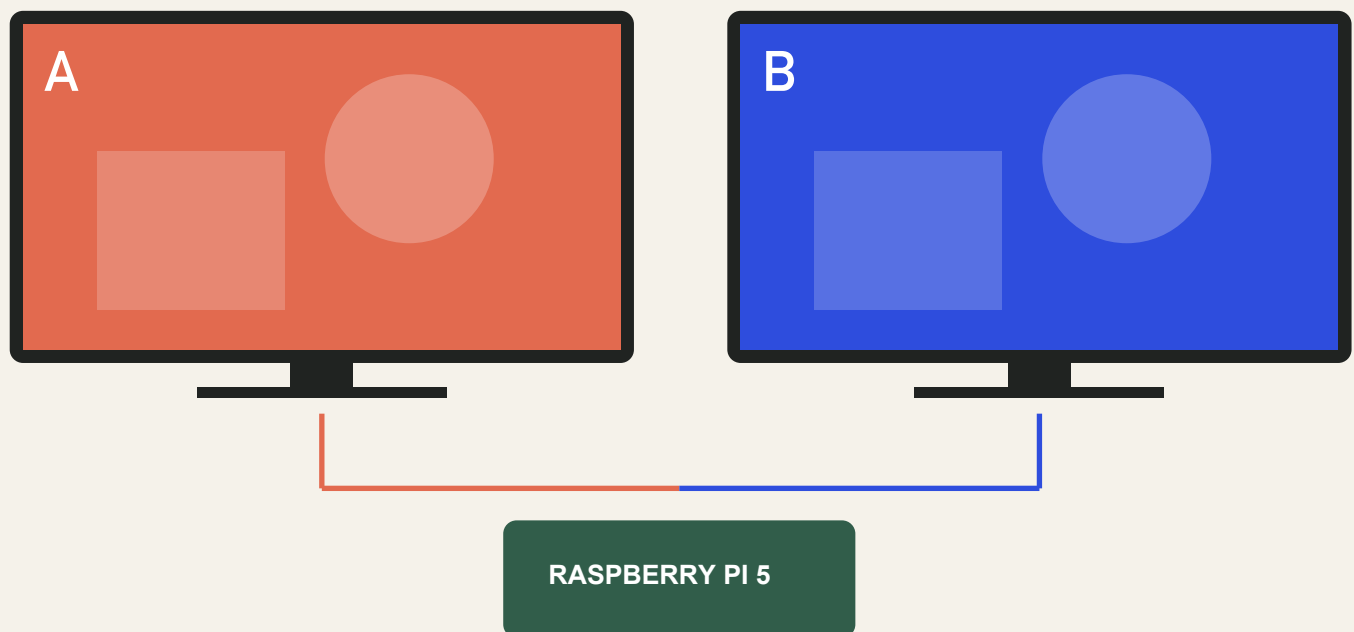


Two screens. One small computer.

Looping video for a graduate exhibition

Raspberry Pi 5

Independent loops + synchronized playback options



Prepare at home. Rehearse with the final files.
Keep the daily crew card beside the installation.

START HERE

Choose how the videos behave.

Decide this before exporting the artwork or ordering equipment.

THE ARTWORK NEEDS	THE PLAYBACK ROUTE
Separate loops A and B can have different lengths. Their timing can drift.	One Pi, two HDMI cables. Follow pages 3-9.
One shared timeline The relationship between A and B must stay fixed.	One combined movie, one Pi HDMI output, plus a cropping video wall controller. See pages 10-11.
Exact frame timing A flash, movement or join must match across screens.	Have an AV technician specify and test synchronized outputs and display delay. See page 11.

WHY TWO PLAYERS DO NOT SYNCHRONIZE

Opening two movies together only gives them a similar starting time. Each player runs its own timeline; a loop or player restart can separate them further. The two-player kit is for independent loops.

Your route through the guide

Prepare	Equipment, exports, SD card and cables	3-6
Set up	Desktop settings, installer, movies and startup	7-9
Adapt	Synchronization and sound	10-12
Run	Rehearsal, daily crew card and troubleshooting	13-15
Reference	Technical review and official sources	16

BEFORE YOU BEGIN

Gather the kit.

Use a dedicated Pi and a fresh SD card. Allow a relaxed afternoon for setup, then an overnight rehearsal.

QTY	ITEM
1	Raspberry Pi 5, 4 GB or 8 GB RAM Either is a sensible starting point for two 1080p movies.
1	Official Raspberry Pi 27 W USB-C power supply Use the appropriate EU plug for the venue.
1	Active Cooler or a Pi 5 case with a fan Fit it before powering on. Leave air space around the enclosure.
1	64 GB or larger microSD card + card reader Choose a reputable A2 card; allow room for the OS and both movies.
2	Powered HDMI screens + two micro-HDMI-to-HDMI cables For independent mode. Each screen needs its own power connection.
1 set	USB keyboard and mouse Keep them available throughout the exhibition.
1	Laptop, USB memory stick and setup-kit ZIP The laptop prepares the card; the USB stick transfers the kit and movies.
A few	Cable labels, spare HDMI cable and a spare prepared SD card Pack the monitor remotes, power strip and any speakers too.

FOR THE SYNCHRONIZED ROUTE

Add a video wall controller that can crop different regions of one input into separate outputs, plus its power and HDMI cables. Have the AV supplier configure it before installation (pages 10-11).

Hardware specifications: Raspberry Pi 5 [1]. The quantities, card capacity and spare kit above are practical recommendations.

PREPARE THE ARTWORK

Export the exhibition files.

Keep the master artwork on your laptop. Make playback copies for the Pi.

SETTING	INDEPENDENT LOOPS
File type	MP4 with H.264 video , 8-bit, standard SDR colour. MP4 is the file container; H.264 is the video format.
Landscape screen	1920 x 1080 pixels . Start with 25 or 30 frames per second.
Portrait screen	1080 x 1920 pixels . Export upright, then rotate the screen in desktop settings (page 7).
Frame rate	Keep the artwork's frame rate where possible. These starting settings need a simultaneous playback test; higher rates add work.
Quality / bitrate	Start around 8-12 Mb/s per 1080p movie . Inspect gradients and fine detail; adjust after testing.
Sound	Silent work: export without audio, or keep the player muted. Sound work: use AAC audio and follow page 12.
Filenames	video-a.mp4 for Screen A. video-b.mp4 for Screen B. Use lowercase names exactly.

PLAYBACK HAS TO BE REHEARSED

The Pi 5 can drive two 4K displays, but H.264 playback uses its CPU. Two 4K movies may overload this setup. Start with 1080p and active cooling; test both final files together. HEVC hardware support does not guarantee acceleration in every player build. [1, 2]

A clean loop: remove accidental black frames and check the last-to-first-frame transition. Audio should also join cleanly. The player repeats the file, but cannot guarantee an invisible or gapless seam.

For artwork that must stay in sync, prepare one combined movie using page 10 instead of these two separate files.

STEP 1 / ON YOUR LAPTOP

Give the Pi its operating system.

The operating system turns the blank board into a small desktop computer.

THIS STEP ERASES THE SELECTED CARD

Use the dedicated SD card and check its size before writing. Unplug other removable drives so you cannot choose them by mistake.

1 Install Raspberry Pi Imager.

Get it from raspberrypi.com/software. Open it and insert the microSD card using the reader.

2 Choose the device, OS and card.

Choose **Raspberry Pi 5**, then **Raspberry Pi OS (64-bit) with Desktop** (Trixie). Select the dedicated microSD card. Use the desktop version; Lite has no desktop. [3, 4]

3 Set your account and location.

Use a simple username such as **exhibit**, choose a password and record it privately. Set the keyboard layout you actually use. For the venue, choose the Netherlands / Amsterdam timezone and Wi-Fi country. Add Wi-Fi for setup. [3]

4 Write, verify and eject.

Proceed through Imager's confirmation. Let writing and verification finish. Eject the card safely, then insert it into the Pi while the Pi is unplugged. [3]

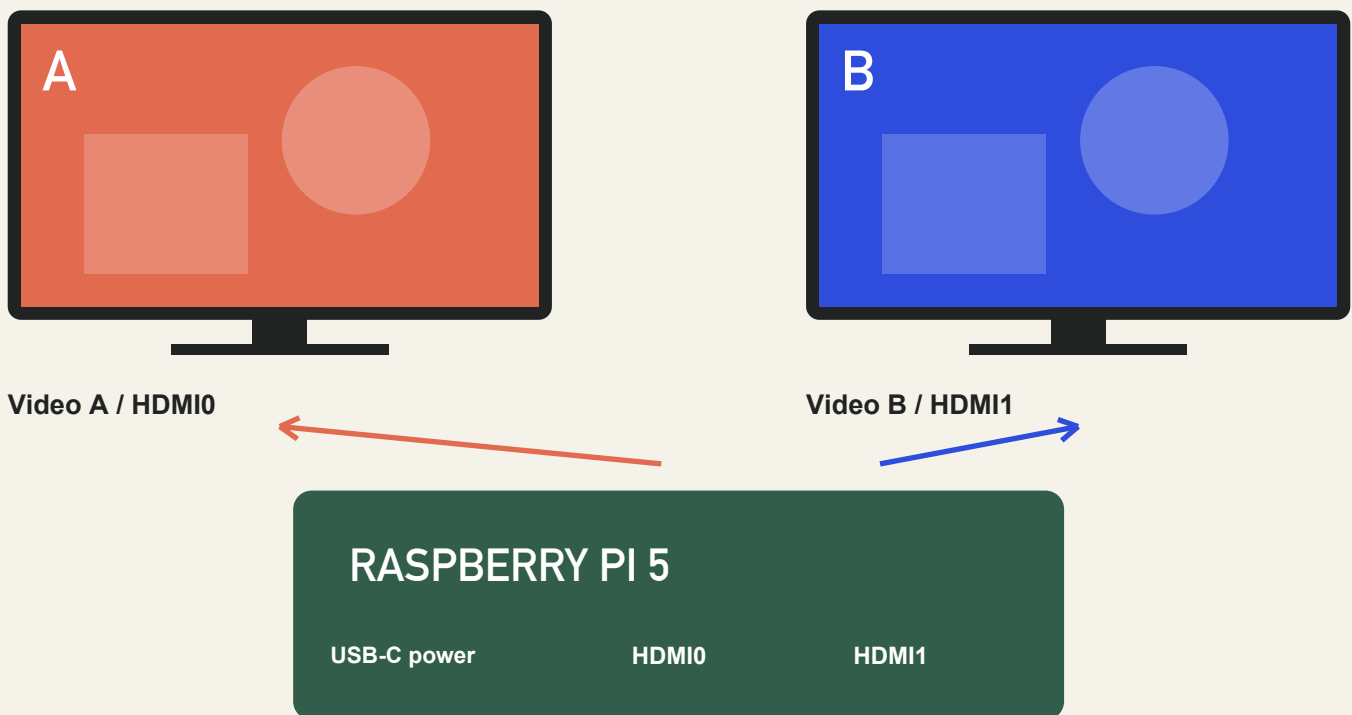
YOU SHOULD NOW HAVE

A card ready to boot the desktop, plus your recorded username and password. The setup scripts use your actual account automatically; it does not have to be named pi.

STEP 2 / AT THE PI

Connect the two screens.

This wiring is for independent loops. Use the controller wiring on page 10 for the shared-timeline route.



Connection diagram only; use the HDMI0 / HDMI1 markings on the actual board.

1 Label both ends of every HDMI cable.

Put **A** on the cable from Pi HDMI0 to Screen A, and **B** on the cable from Pi HDMI1 to Screen B. Insert the keyboard, mouse and prepared SD card. [3]

2 Turn the screens on first.

Use each monitor's remote or buttons to choose the HDMI input you plugged into. Then connect the Pi's USB-C power supply. Finish any first-boot prompts. [3]

YOU SHOULD SEE

A desktop on the screens. If a screen says No Signal, first check its selected input, its power and the cable. A splitter is unnecessary for independent videos.

STEP 3 / ON THE PI DESKTOP

Set the screens and startup.

Open the Raspberry Pi menu, then Preferences > Control Centre. Menu wording can vary slightly with updates.

WHERE	WHAT TO SET
Screens	Enable both displays and arrange them side by side . Choose an extended desktop, with mirroring off. Start at 1920 x 1080 on each landscape monitor. [5]
Portrait work	In Screens, rotate the relevant display 90 or 270 degrees so the desktop is upright. Keep its native 1920 x 1080 mode; rotation makes the usable canvas 1080 x 1920. [5]
System	Choose Boot: To desktop and turn Desktop Auto Login ON . Save or close the settings. [5]
Display	Turn Screen Blanking OFF , then close the settings. [5]
Each monitor	In its own on-screen menu, disable sleep timers, automatic power-off and input switching. Use a fit-to-screen / Just Scan mode if edges are cropped.

Check the display names

Open Terminal with **Ctrl + Alt + T**. Type this, then press Enter:

```
wlr-randr
```

You should see two outputs. They are usually **HDMI-A-1** (HDMI0) and **HDMI-A-2** (HDMI1). Confirm which screen is which in Screens settings; the installer will ask for their names. Do not assume left/right position proves the port mapping.

YOU SHOULD NOW HAVE

Two desktop areas, correct orientation, automatic login and no idle screen blanking. The video kit preserves picture proportions, adding black bars where needed.

STEP 4 / ON THE PI

Install the playback kit.

This replaces the long configuration blocks in the original conversation with a guided setup.

1 Unzip and copy the kit.

Extract **Exhibition-Setup-Kit.zip** on your laptop. Copy the **exhibit-kit** folder to a USB stick. On the Pi, open File Manager and copy that whole folder into **Home**. Its name must stay **exhibit-kit**.

2 Open Terminal on the Pi.

Press **Ctrl + Alt + T**. Type each line below and press Enter after each one. The **~** symbol means your Home folder.

```
cd ~/exhibit-kit
bash install.sh
```

The Pi needs internet for this step. If asked, type your Pi password and press Enter; the password stays invisible. Let the installation finish. If it stops with an error, keep the message for your helper.

3 Answer the short setup questions.

For two direct HDMI cables, choose **1: INDEPENDENT**. Select Screen A's output, then Screen B's. Choose **SILENT** or sound from Video A. For the controller route, choose **2: SHARED TIMELINE** and its input output (page 11).

WAIT FOR SETUP COMPLETE

The kit creates a folder called **exhibit** in Home, plus **Start Exhibit** and **Stop Exhibit** controls. It backs up the desktop configuration, assigns movies to screens and adds automatic player restarts. It does not start the movies yet.

Run the kit as your ordinary desktop user, without putting **sudo** before **bash install.sh**. The kit requests **sudo** only for package installation.

STEP 5 / ON THE PI

Copy the movies. Press play.

The movies live on the Pi's SD card. A USB stick and internet are not needed during playback.

1 Copy into Home > exhibit.

Copy **video-a.mp4** and **video-b.mp4** from the USB stick into the **exhibit** folder. Check spelling and lowercase letters. Shared mode uses only **shared.mp4**. Wait until copying finishes, then eject the USB stick.

2 Start the exhibition.

Double-click **Start Exhibit** on the desktop. If prompted, choose Execute / Allow Launching. You can also open it from the application menu or use the command below.

```
python3 ~/exhibit/control.py start
```

YOU SHOULD SEE

Video A filling Screen A and Video B filling Screen B, with no player buttons or mouse pointer. Each repeats. If the artwork is narrower than the screen, black bars preserve its proportions.

3 Reboot to prove automatic startup.

Stop the videos using page 14, then choose **Restart** from the Pi menu. Leave both screens on. After the desktop appears, the movies should start by themselves. Allow up to about two minutes on a fresh boot.

4 Check the work, not just the computer.

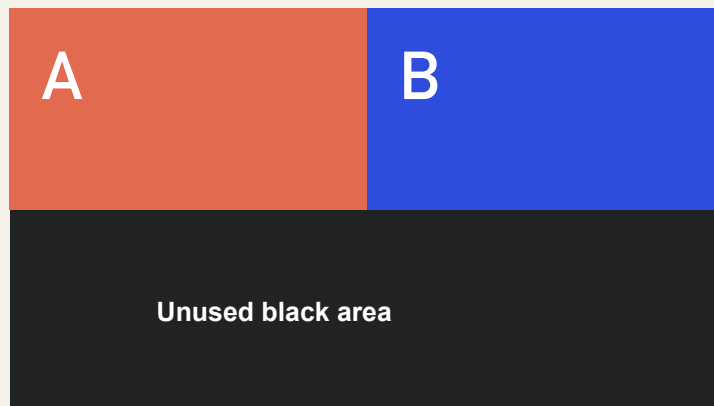
Watch the beginning and at least two loop joins. Confirm the correct artwork, orientation, edge framing and sound. Then run the overnight rehearsal on page 13.

The normal player keys are disabled to avoid accidental pauses. Use the Start / Stop controls instead of q, Escape or the space bar.

SYNCHRONIZED OPTION / THE ARTWORK

Put both pictures in one movie.

This suggested route uses one playback timeline and a separate device to crop it into two screen images.



One 3840 x 2160 frame

A: top left, 1920 x 1080

B: top right, 1920 x 1080

Both share one timeline.

1 Build a combined composition in your video editor.

Make a **3840 x 2160** canvas. Place A at the top left and B at the top right, each at 1920 x 1080 with no gap. Leave the bottom half black. Align both on one timeline and give the composition one agreed loop length.

2 Export one file named **shared.mp4**.

For this 4K canvas, try **HEVC / H.265, 8-bit SDR, 25 or 30 fps**. Test the actual file on the Pi. HEVC acceleration depends on the installed playback stack; this kit's automatic hardware request may fall back to CPU decoding. [1, 2, 6]

3 Send one HDMI signal to a cropping controller.

Connect Pi **HDMI0 to the controller input**. Connect controller outputs 1 and 2 to Screens A and B. Pi HDMI1 is unused during the show. Ask the technician to crop the two top regions separately. [9]

A SPLITTER CANNOT DO THIS JOB

An ordinary HDMI splitter duplicates the entire frame. You need a **video wall controller with independent crop regions**, preconfigured by the supplier. A Datapath Fx4-HDR is one documented example; arrange a rental or suitable equivalent with the venue. [9]

SYNCHRONIZED OPTION / AV HANDOFF

Give this page to the technician.

The controller and displays need their own rehearsal. One shared movie preserves the content timing; display refresh and processing can still create a visible offset.

CHECK	REQUEST
Input	Pi HDMI0, 3840 x 2160 at 25 or 30 Hz . Use a controller that accepts this timing and stores its mapping through power cycles.
Crop A	Source rectangle x=0, y=0, width=1920, height=1080 to output 1.
Crop B	Source rectangle x=1920, y=0, width=1920, height=1080 to output 2.
Outputs	Set both to 1920 x 1080 at the same suitable refresh rate, without cropping the artwork. Configure rotation / scaling in the controller for portrait screens.
Timing test	Use a temporary numbered-frame or flashing-marker version of shared.mp4. Check both screens at startup, loop joins and after an hour. Match monitor picture modes and processing delay.

Use the Pi kit in shared mode

With the controller connected, install the kit (page 8), select **SHARED TIMELINE** and select the Pi output that feeds it. In Screens, set that output to the agreed 4K timing. Copy **shared.mp4** into Home > exhibit, then use the same Start / Stop controls. Only one player runs. Test controller power-on before Pi startup.

IF THE 4K MOVIE STUTTERS

A lower-load proof of concept uses a **1920 x 1080** canvas with A and B at **960 x 540** in the two top corners. Change the crop regions to match; the controller upscales them. Detail is reduced. If that is unacceptable, use an AV playback computer for the same combined-file route.

For exact frame timing: ask the AV team to verify output timing and screen latency. Another route is two compatible BrightSign players on wired Ethernet with Enhanced Sync / PTP. Verify the models, formats and resynchronization at every loop. [10]

OPTIONAL / SOUND

Route the sound.

A movie's screen position does not automatically choose its speaker output.

YOUR WORK	SETUP
Silent artwork	Choose SILENT during setup. Both videos stay muted.
One soundtrack	Choose SOUND. Video A, or shared.mp4, uses the desktop default sound output. Select it in the speaker menu and set the volume.
Separate sound	Use the sound helper below. Choose a distinct HDMI or USB audio output for each movie. A shared output mixes both sounds.

Open Terminal on the Pi and run:

```
python3 ~/exhibit/audio.py
```

The helper stops playback and lists sound devices. Choose **0** for silence, or the number of the output you want for each movie. Click **Start Exhibit** again and listen at each screen. If the list is empty, connect or enable the audio device and try again.

Test by listening: use a temporary clip that says "A" or "B". Confirm the physical speakers. Recheck after changing monitors, HDMI cables or USB audio devices.

A few practical choices

- ☐ The Pi 5 has no built-in 3.5 mm audio socket. Use HDMI audio, a USB audio device or speakers connected to the monitor. [1, 3]
- ☐ Avoid Bluetooth for timing-critical sound. Keep the chosen speakers and audio connections consistent during rehearsal and installation.
- ☐ For the shared movie, make one deliberate soundtrack or mix in your video editor. Ask the AV supplier how the controller handles sound.

Audio selection and device listing use mpv's documented options [6]. The kit uses named devices when you select them.

BEFORE YOU TRAVEL

Rehearse the actual exhibition.

Use the final movies, final monitors, cables, sound, enclosure and power supply. Record any change and repeat the affected checks.

- ☐ **First look:** the correct artwork is on each screen; portrait work is upright; no unwanted cropping, interface or mouse pointer is visible.
- ☐ **Motion:** both pictures play smoothly together. Watch slow pans, texture, fades and gradients as well as fast movement.
- ☐ **Loop:** watch several end-to-start transitions. Check for black frames, a frozen image, a sound click or a noticeable pause.
- ☐ **Overnight:** run at least 8-12 hours. Check again in the morning. The screens should still be on and the Pi enclosure should have clear ventilation.
- ☐ **Cold startup:** shut down properly, disconnect Pi power, reconnect with the screens already on. Repeat this three times. Each boot should begin playback without a login or button press.
- ☐ **Recovery:** use Stop Exhibit and Start Exhibit. If a player exits, the kit should relaunch it after about five seconds. A helper can test a simulated crash with the command below.

```
systemctl --user kill -s SIGKILL exhibit-a.service
```

In shared mode, replace exhibit-a.service with exhibit-shared.service. This tests a player crash. It does not test a frozen player or a loss of electricity.

MAKE THE INSTALLATION RECOVERABLE

After the final rehearsal, have a helper clone the working SD card to an equal or larger card and boot-test the clone. Bring it in a labeled case, plus the original movies and the setup kit. A blank spare card will not restore the installation.

Power-loss recovery is expected from desktop auto login and autostart, but only a real venue test can confirm the full chain. If needed, ask the technician to test it with a backup card available. Avoid routine power cuts.

PRINT AND KEEP BESIDE THE WORK

Daily crew card.

Use this page after the setup has passed its full rehearsal.

OPEN THE EXHIBITION

1. Turn on both screens.

Choose the correct HDMI inputs. In shared mode, turn on the controller too.

2. Power the Pi.

Reconnect its power or briefly press the Pi's power button if it was shut down while still plugged in.

3. Wait for playback.

Allow about two minutes. Both pictures should play automatically.

4. Check the work.

Correct videos, orientation, sound and movement. Keep the vents clear.

Terminal commands:

```
# Stop playback
python3 ~/exhibit/control.py stop
# Restart the movies without rebooting
python3 ~/exhibit/control.py start
```

CLOSE THE EXHIBITION

1. Connect the keyboard.

Press Ctrl + Alt + T.

2. Stop playback.

Type the Stop command below and press Enter.

3. Shut down the Pi.

Use the Pi menu > Logout / Shutdown, then choose Shut Down.

4. Wait.

After the screen loses its signal, allow another 10 seconds before disconnecting Pi power. Then switch off the screens.

IF A SCREEN IS BLANK OR FROZEN

Check its power and HDMI input first. Restart playback with the command above. If it remains stuck, shut down the Pi properly and start again with the screens already on. Contact the helper if this repeats. Never remove the SD card while the Pi is powered.

HELPER NAME / PHONE

WHEN SOMETHING GOES WRONG

Start with the visible symptom.

Use the simple checks first. Keep any error message for your helper.

SYMPTOM	FIRST ACTION
No Signal	Check monitor power, selected HDMI input and both cable ends. Start with the monitors on before the Pi. Try the spare cable.
Desktop, no movies	Check Home > exhibit for the exact filenames. Click Start Exhibit. Check Desktop Auto Login and reboot. Setup errors are in startup.log.
Videos on wrong screens	Stop playback. Confirm A / B cable labels and output names. Run the installer again and correct the Screen A / B choices; it keeps your movies.
Both screens look identical	For independent mode, turn desktop mirroring off. Confirm that the two files are different. An ordinary splitter cannot provide two pictures.
Small picture / black bars	The player preserves proportions. Match the export to the screen shape. Check screen rotation and the monitor's fit-to-screen setting.
Stops after a few minutes	Check Pi screen blanking and the monitor's own sleep / power-off settings. A still-running frozen player needs a manual restart.
Jerky video / hot Pi	Clear vents and check the fan and official power supply. Try the 1080p export settings. Keep both exact files running during your test.
Wrong or missing sound	Check speaker power, mute and volume. Run the sound helper (page 12). Screen assignment and sound assignment are separate.

For your helper

```
wlr-randr
journalctl --user -u exhibit-a -u exhibit-b -u exhibit-shared -n 80 --no-pager
vcgencmd get_throttled
```

Copyable commands are in START-HERE.txt. A helper should interpret get_throttled.

What was checked and improved.

The original two-player approach is suitable for independent loops. This guide adds exhibition operation and a separate route for synchronized artwork.

Kept	Pi 5, an extended desktop, mpv loops and desktop auto login. The core design is appropriate for separate movies.
Made more reliable	The kit matches fixed Wayland app IDs, moves each window before fullscreen, backs up existing XML and restarts each player independently. [7, 8]
Clarified	Two 4K display outputs do not promise two smooth 4K decodes. HDMI video placement does not route audio. Two separate players do not maintain sync.
Added	Start / Stop controls, named audio devices, a combined-file controller option, rehearsal checklist and daily crew card.

Verification: official specifications and configuration documentation were reviewed in October 2026. The included scripts passed syntax and configuration-logic checks. **No physical Pi or venue screens were available for a playback test.** The final overnight rehearsal remains necessary.

Official references

- [1] [Raspberry Pi 5 specifications](#)
- [2] [Raspberry Pi: H.264 / HEVC software environment](#)
- [3] [Raspberry Pi: getting started and Imager](#)
- [4] [Raspberry Pi OS: current desktop / Trixie downloads](#)
- [5] [Raspberry Pi: screens, rotation, login and blanking](#)
- [6] [mpv: playback, looping, Wayland and audio options](#)
- [7] [labwc: window rules and app identifiers](#)
- [8] [labwc: MoveToOutput and ToggleFullscreen](#)
- [9] [Datapath Fx4-HDR: cropping / multi-output controller](#)
- [10] [BrightSign: shared-clock synchronization architecture](#)
- [11] [systemd: automatic service restart behaviour](#)

Numbers in the guide refer to these clickable links. Controller crop coordinates and export settings are proposed configurations to rehearse, rather than manufacturer performance guarantees.